

Challenge Set 1: UDP Analysis

Samples to use: DNS + NTP

- 1. Open the DNS sample.
 - o Locate the UDP header.
 - o Write down the Source Port and Destination Port.
 - o What is the Destination Port number and which well-known service does it represent?

Source Port: 59198
Destination Port: 53
Destination Port 53 represents the DNS (Domain Name System) service.

192.168.56.11 → 194.190.124.17 DNS Standard query 0xe4a3 A ns2.highsecure.ru OPT

⬆️ Ask AI about this packet

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
52	54	00	7E	42	A5	52	54	00	A6	11	53	08	00	45	00
00	4A	4A	48	00	00	3F	11	F9	D7	C0	A8	38	0B	C2	BE
7C	11	E7	3E	00	35	00	36	37	CB	E4	A3	00	10	00	01
00	00	00	00	00	01	03	6E	73	32	0A	68	69	67	68	73
65	63	75	72	65	02	72	75	00	00	01	00	01	00	00	29
10	00	00	00	80	00	00	00								

4 protocols in packet (88 bytes):

EthernetIPv4UDPDNSX

User Datagram Protocol
Source Port: 59198

Field offset: [34-35]
Field length: 2
Byte offset: 34

User Datagram Protocol
Destination Port: 53

Field offset: [36-37]
Field length: 2
Byte offset: 36

- 2. Still in the DNS sample:
 - o Find the UDP Length and Checksum fields.
 - o Does this packet have any sequence number or acknowledgement number? What does this tell you about UDP?

User Datagram Protocol
Length: 54

Field offset: [38-39]
Field length: 2
Byte offset: 38

00	36	37	CB	E4
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User Datagram Protocol
Checksum: 0x37cb [correct] (matches partial checksum, not 0x647a, likely caused by "UDP checksum offload")

Field offset: [40-41]
Field length: 2
Byte offset: 40

UDP Length: 54

UDP Checksum: 0x37CB

Sequence Number: Not present

Acknowledgement Number: Not present

Conclusion: UDP does not use sequence numbers or acknowledgement numbers. This shows that UDP is a connectionless transport protocol and does not provide TCP-style reliability mechanisms.

3. Open the NTP sample.

- o Compare its UDP header with the DNS sample.
- o Are the header fields the same size?
- o Why is UDP suitable for both DNS and NTP?

The UDP headers of DNS and NTP contain the same four fields: Source Port, Destination Port, Length, and Checksum.

The UDP header has a fixed size of 8 bytes.

UDP is suitable for both DNS and NTP because it has low overhead and provides fast, connectionless communication. Both applications can use UDP port numbers to identify the required service.

216.27.185.42 → 192.168.50.50 NTP NTP Version 3, symmetric passive [+. Ask AI about this packet](#)

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
00	D0	59	6C	40	4E	00	0C	41	82	B2	53	08	00	45	00
00	4C	A2	22	40	00	32	11	22	5E	D8	1B	B9	2A	C0	A8
32	32	00	7B	00	7B	00	38	70	E1	1A	02	0A	EE	00	00
07	A4	00	00	0B	A3	A4	43	3E	C2	C5	02	01	81	E5	79
18	19	C5	02	04	EC	EC	42	EE	92	C5	02	04	EB	D9	5E
8D	54	C5	02	04	EB	D9	69	B1	74						

4 protocols in packet (90 bytes):

Ethernet IPv4 UDP NTP X

Challenge Set 2: TCP Analysis

Samples to use: FTP + MySQL

4. Open the FTP sample.

- o Expand the TCP header.
- o List the following fields with their values:
 - ♣ Source Port
 - ♣ Destination Port

- ♣ Sequence Number
- ♣ Acknowledgement Number
- ♣ Flags (which flags are set?)

Source Port: 21

Destination Port: 62014

Sequence Number: 0x836F4FFE

Acknowledgement Number: 0x04846EDA

Flags: PSH and ACK

192.168.1.231 → 192.168.1.182 FTP Response: 220 ProFTPD 1.3.0a Server (ProFTPD Anonymous Server) [192.168.1.231]

[+ Ask AI about this packet](#)

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
00	17	F2	2E	52	3B	00	13	20	9F	04	2F	08	00	45	10
00	7A	E6	5E	40	00	40	06	CF	21	C0	A8	01	E7	C0	A8
01	B6	00	15	F2	3E	83	6F	4F	FE	04	84	6E	DA	80	18
05	A8	99	67	00	00	01	01	08	0A	3C	F6	A9	AA	1E	8D
47	F6	32	32	30	20	50	72	6F	46	54	50	44	20	31	2E
33	2E	30	61	20	53	65	72	76	65	72	20	28	50	72	6F
46	54	50	44	20	41	6E	6F	6E	79	6D	6F	75	73	20	53
65	72	76	65	72	29	20	5B	31	39	32	2E	31	36	38	2E
31	2E	32	33	31	5D	0D	0A								

4 protocols in packet (136 bytes):

Ethernet IPv4 TCP FTP X

Transmission Control Protocol Source Port: 21 Field offset: [34-35] Field length: 2 Byte offset: 34	Transmission Control Protocol Destination Port: 62014 Field offset: [36-37] Field length: 2 Byte offset: 36
Transmission Control Protocol Sequence Number: 0x836f4ffe Field offset: [38-41] Field length: 4 Byte offset: 38	Transmission Control Protocol Acknowledgment Number: 0x04846eda Field offset: [42-45] Field length: 4 Byte offset: 42

5. In the same FTP sample:

- o What is the 4-tuple that uniquely identifies this TCP socket?
- o Explain how this 4-tuple helps in demultiplexing.

The 4-tuple is:

(192.168.1.231, 21, 192.168.1.182, 62014)

The 4-tuple consists of the source IP address, source port, destination IP address, and destination port. It uniquely identifies a TCP connection and helps the operating system perform demultiplexing by delivering each incoming packet to the correct TCP socket/application.

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Internet Protocol Version 4

Src: 192.168.1.231
Dst: 192.168.1.182
Length: 20 bytes
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6. Open the MySQL sample.

o Compare its TCP ports with the FTP sample.

o Are the Destination Ports the same or different?

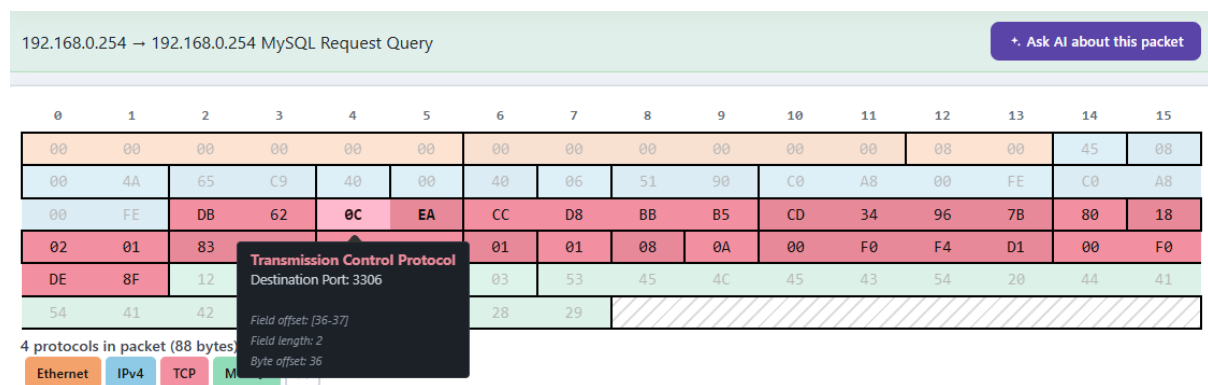
o How does the operating system know which application should receive each packet?
(Demultiplexing)

FTP Destination Port: 62014

MySQL Destination Port: 3306

The Destination Ports are different.

The operating system uses the IP addresses and port numbers of each TCP connection to perform demultiplexing. This allows it to identify the correct socket and deliver each packet to the appropriate application.



Challenge Set 3: TCP vs UDP Comparison

Samples to use: DNS (UDP) + FTP (TCP)

7. Place the DNS and FTP samples side-by-side (open one after another).

o Which protocol has a larger header?

o Which protocol provides reliability features (sequence numbers, flags, window)?

o Which protocol is connection-oriented and which is connectionless? Give evidence

TCP has a larger header than UDP. The minimum TCP header size is 20 bytes, while the UDP header is 8 bytes.

TCP provides reliability features such as sequence numbers, acknowledgement numbers, flags, and window size. UDP does not provide sequence numbers or acknowledgement numbers.

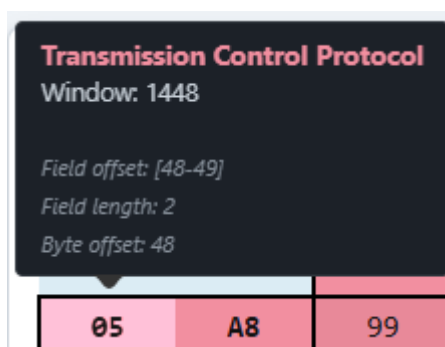
TCP is connection-oriented, while UDP is connectionless. The FTP sample shows TCP sequence and acknowledgement numbers, whereas the DNS sample using UDP does not contain these fields.

8. From the FTP sample: o Point out any fields that show TCP's duty of providing reliability and flow control.

The FTP TCP header contains several fields that provide reliability and flow control:

- **Sequence Number:** 0x836F4FFE — helps maintain the correct order of data.
- **Acknowledgement Number:** 0x04846EDA — acknowledges received data.
- **ACK Flag:** Set — indicates that acknowledgement information is present.
- **Window Size:** 1448 — provides flow control by indicating the amount of data the receiver can accept.

Therefore, Sequence Number and Acknowledgement Number support reliability, while Window Size provides flow control.



Challenge Set 4: Modern Transport

Sample to use: QUIC

10. Open the QUIC sample.

o Which underlying Transport protocol is QUIC using? (Look carefully)

o What is the Destination Port?

o How is QUIC different from classic TCP even though it runs over UDP?

Underlying Transport Protocol: UDP

Destination Port:443

QUIC runs over UDP but provides its own transport features, including reliability and connection management. Unlike classic TCP, QUIC does not use the TCP protocol and header; it uses UDP as its underlying transport.

2001:db8:1::1 → 2606:4700:10::6816:826 QUIC Handshake

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 Ask AI about this packet

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
00	03	2D	87	59	70	F0	18	98	E8	7A	F3	86	DD	60	03
0E	00	00	3D	11	40	20	01	0D	B8	00	01	00	00	00	00
00	00	00	00	00	01	26	06	47	00	00	10	00	00	00	00
00	00	68	16	08	26	C4	68	01	BB	00	3D	F6	CF	E4	00
00	00	01	14	01	30	DF	C5	A0	47	E6	AC	D2	30	B5	C5
E0	47	CE	D9	B0	A6	BB	F0	00	40	18	62	62	7A	4B	50
9A	40	68	12	B7	1D	F9	BE	5A	8C	E4	F8	95	DD	26	6E
54	56	7D													

4 protocols in packet (115 bytes):

EthernetIPv6UDPQUICX

User Datagram Protocol

Destination Port: 443

Field offset: [56-57]

Field length: 2

Byte offset: 56

C4	68	01	BB	00
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